

The Value of Sharing Rebalancing Information on YouBike Users’ Waiting Cost: A Data-Driven Time-Step Simulation

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<https://github.com/CN-Team4/final>

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Abstract

Bike-sharing supply is highly uneven in space and time; during peak hours, popular stations frequently run out of bikes. Operators rebalance the fleet with dispatch vehicles, but the location and timing of the dispatch fleet are known only to the operator, not to users. This paper asks a question orthogonal to dispatch *optimization*: under a fixed, resource-constrained dispatch operation, does simply *revealing* the current dispatch-vehicle locations to users reduce their travel cost? We build a data-driven, time-step simulation of peak-hour shortage over a 76-station cluster around National Taiwan University (NTU) and compare two scenarios—*with* versus *without* dispatch information—that share an *identical* dispatch supply (location and capacity); the only difference is whether users know where the dispatch bikes are. The main metric is the difference in mean arrival time between the two scenarios, which isolates the value of the information itself. All key quantities (hourly departure distribution, dispatch footprint, shortage rate, dock-fullness) are estimated from 43.9 GB of real-time station-status data (Sept. 2025, weekdays) and monthly origin–destination statistics; the only purely behavioral assumption is a detour-tolerance ratio τ , for which we provide a sensitivity analysis. At NTU’s empirically measured shortage level ($\approx 32\%$) and across a physically plausible dispatch capacity (50–200 bikes per peak window), revealing dispatch locations lowers mean arrival time by **1.8–2.4 minutes (15–22%)**; in the 10–20% shortage range it saves **2.4–3.2 minutes (22–31%)**. The conclusion is robust to τ . We further find that the operator’s actual main dispatch hub (MRT Gongguan, at the south-west edge) is sub-optimal for the campus shortage population, whose residual unmet demand concentrates in the eastern dormitory/department area.

Keywords: bike sharing, information disclosure, rebalancing, agent-based simulation, network service design.

1 Introduction

1.1 Problem and Motivation

Public bike-sharing systems such as YouBike exhibit strong spatio-temporal demand imbalance, so that during peak hours certain stations are emptied of bikes while others fill up. Operators counter this with *dispatch vehicles* that move bikes between stations. However, where and when these vehicles deposit bikes is internal operational information; a user who cannot find a bike nearby has no way of knowing that a freshly replenished station exists a short walk away, and therefore resorts to guessing. We ask:

If the real-time location of the (already existing) dispatch bikes were disclosed to users—e.g., via an app layer—would it shorten their time to destination?

This is a network-service-design question about the effect of *information disclosure* on system efficiency, deliberately distinct from how to *optimize* the dispatch policy.

1.2 Our Angle

Prior work mostly *predicts* station occupancy or *optimizes* the dispatch policy. Yet field reports indicate that real dispatch is bottlenecked by limited vehicles, manpower, and turnaround speed, so a better policy does not automatically translate into better service. We therefore hold the (limited) dispatch supply fixed and measure the marginal benefit of *merely revealing the existing dispatch information*. The study follows three principles:

- (1) **Information is the only manipulated variable.** Both scenarios use the same dispatch locations, capacities, and counts; they differ only in whether the user knows them. Biases in the data (existing rebalancing, censored demand) are common-mode and cancel in the cost difference.
- (2) **No supply is added.** We quantify the marginal benefit of disclosing *existing* dispatch locations, not of adding vehicles.
- (3) **Replace assumptions with data.** Dispatch footprint, shortage rate, departure distribution, and dock-fullness are all estimated from real data; the only behavioral assumption is the detour tolerance τ , which we sweep.

1.3 Contributions

(1) A data-driven peak-hour shortage simulation of the NTU YouBike cluster from two open datasets (44 GB real-time dynamics + monthly OD). (2) A quantification of the *time value of dispatch-information disclosure*, and a two-dimensional characterization (shortage level $\times$ dispatch quantity) of when it helps most. (3) An open-source, reproducible implementation with interactive maps and an agent-movement animation.

2 Related Work

Prior studies cluster into prediction, dispatch optimization, and information services.

- **Occupancy prediction.** Yang (2022) [1] models future available-bike counts per station from history, temporal features, and weather, exposed as a web service. *Limitation:* answers “how many bikes a station may have,” not rebalancing optimization.
- **Dispatch optimization around NTU.** Chou (2025) [2] analyzes rental/return hotspots and ride times and compares dispatch strategies (greedy, a survival-analysis SurvivalALG, and an offline near-optimal method). *Limitation:* the focus is “how the operator should dispatch.”
- **Zoning and routing optimization.** Ko (2015) [3] integrates depot location, zoning, vehicle count, and routing cost and solves it with simulated annealing. *Limitation:* suited to operator mid/long-term planning with well-specified demand and routes.
- **User-facing information services.** Chen et al. (2022) [4] propose app features (reservation, LED displays, back-end statistics). *Limitation:* argues users need better information but does not quantify how much time a given piece of information saves.
- **Operational constraints.** Su (2023) [5] documents that the operator’s dispatch vehicles, manpower, throughput, and replenishment speed are tight, so even knowing which stations run dry does not guarantee timely replenishment.

Gap. Existing work optimizes policy or predicts occupancy; none quantifies the time benefit of disclosing *existing* dispatch information to users under a fixed, constrained dispatch operation—the question we address.

3 Data

The dynamics file is header-less; column meanings follow the official YouBike API schema and were verified against value relationships (e.g., `sbi + bemp  $\approx$  tot`). We restrict to September 2025 weekdays (excluding the August summer break and weekends) to align with the weekday OD data and the in-semester regime. Processing uses streaming reads on a 144-core machine (≈ 60 s per full pass); both datasets are fetched by `fetch_data.sh`.

Table 1: Datasets.

Source	Period	Use
YouBike 2.0 real-time station dynamics (≈ 43.9 GB)	Sept. 2025, weekdays	Departure distribution $P^i(t)$; dispatch footprint; shortage rate; dock-fullness (from available bikes sbi and empty docks bemp)
Taipei YouBike OD statistics (data.taipei)	Dec. 2025, weekdays	Monthly OD demand D_m^{ij} ; station graph

4 System Model and Assumptions

4.1 Station Cluster (76 stations)

Instead of an arbitrary radius, the cluster follows YouBike’s official zoning: *NTU zone* $\cup$ stations named “NTU” (including the peripheral dormitory belt) $\cup$ the Gongguan exits (goal) – the downtown NTU-Hospital stations = **76 stations**. Stations form a complete graph; distances are Manhattan; riding speed is $5\times$ walking ($V_{\text{walk}} = 1.25$, $V_{\text{bike}} = 6.25$ m/s).

4.2 Dispatch Resource

Location/distribution is detected from data: a single-step jump of ≥ 10 in a station’s available bikes is attributed to a dispatch truck (ordinary returns add one at a time). Within the cluster, **37 stations** receive refills; their relative shares come from the data. **Total quantity** is hard to measure: the detected total (≈ 213 bikes/window) may be inflated by return bursts, and the campus capacity is roughly $2 \text{ trucks} \times 25$ plus on-site *decompression* (bikes stored beside a station and re-racked) and possibly multiple trips. We therefore **anchor the total at a physical mid-estimate of 100 bikes per peak window** and treat it as an *uncertain parameter*, sweeping a scale s to $25/50/100/200/400$; **50–200 is the plausible band**. Both scenarios use the identical footprint and capacity.

4.3 Population and Behavior

The population is the *shortage group* (size αD_t), spatially weighted by each station’s measured peak **sbi** = 0 rate. Users are rational shortest-path agents and *do not wait* (excluding a waiting branch isolates the value of *knowing the location*, not the side-effect of acting instead of waiting). Cars, ride-sharing, and reverse-direction search are not modeled.

5 Methodology

5.1 Demand Estimation

$$D_d^{ij} = D_m^{ij}/N_d, \quad D_t^{ij} = D_d^{ij} P^i(t), \quad \text{shortage} = \alpha D_t^{ij} w_i, \quad (1)$$

where N_d is the weekday count, $P^i(t)$ the per-station hourly departure share, and w_i a normalized shortage weight (so the total shortage is $\alpha \sum D_t$ while its spatial pattern matches the measured shortage). The shortage ratio is $\alpha/(1 + \alpha)$; α is the swept variable.

5.2 Empirical Estimation

$P^i(t)$: between consecutive status changes, a drop in **sbi** counts as a borrow; borrows are aggregated by hour and normalized per station (1,651 stations). The simulation uses each origin’s own $P^i(t)$ (the cluster curve is shown only for illustration); the NTU curve is multi-peak ($\approx 09:00, 13:00, 17:00$) with peak window 17:00–19:00. **Shortage rate:** the peak **sbi** = 0 time fraction, demand-weighted $f \approx 32\%$, i.e. $\alpha \approx 0.48$. **Dock-fullness:** peak fraction with **bemp** ≤ 2 ($\approx 80\text{--}90\%$ at the Gongguan exits), used for return cost.

5.3 User Decision and Cost (no waiting)

Without information: walk to the single nearest station; if it is a dispatch station with capacity, ride to the destination; otherwise walk on. **With information:** walk to the nearest dispatch station within tolerance $\tau \cdot \text{dist}(i, j)$ that still has capacity and ride; if none, walk. Let nv be the station nearest to destination j and $f_{\text{full},j}$ its measured fullness; a rider’s expected return detour is

$$\text{ret}_j = f_{\text{full},j}(\text{dist}(j, nv)/V_{\text{bike}} + \text{dist}(nv, j)/V_{\text{walk}}). \quad (2)$$

The resulting cost has three cases. (i) *Reach a bike at station x and ride* (with-info success at $x=g$, or without-info success when the nearest station m is a dispatch station): $\text{dist}(i, x)/V_{\text{walk}} + \text{dist}(x, j)/V_{\text{bike}} + \text{ret}_j$. (ii) *With-info give-up* (no reachable dispatch): walk straight to the destination, $\text{dist}(i, j)/V_{\text{walk}}$. (iii) *Without-info miss* (the nearest station m is not a dispatch station): walk via m , $\text{dist}(i, m)/V_{\text{walk}} + \text{dist}(m, j)/V_{\text{walk}}$.

5.4 Simulation

A time-step simulation allocates each dispatch station’s capacity first-come (by arrival order). Each (scenario, parameter) combination runs 30 Monte-Carlo repetitions (Poisson shortage counts); we report means with 95% confidence intervals. Both scenarios use the same agents and capacities.

6 Experimental Setup

Exp. 1: fix the real footprint ($s=1$, $\tau=0.5$) and sweep α . **2-D sweep:** $\alpha \times s$ (shortage $\times$ dispatch quantity). **τ sensitivity:** sweep $\tau \times \alpha$. **Appendix:** concentrate the total capacity at a single station and sweep all 76 to find the best location. **Baseline:** without information. Environment: Python 3.14, polars, NumPy, Matplotlib, Folium (Leaflet), contextily.

7 Results

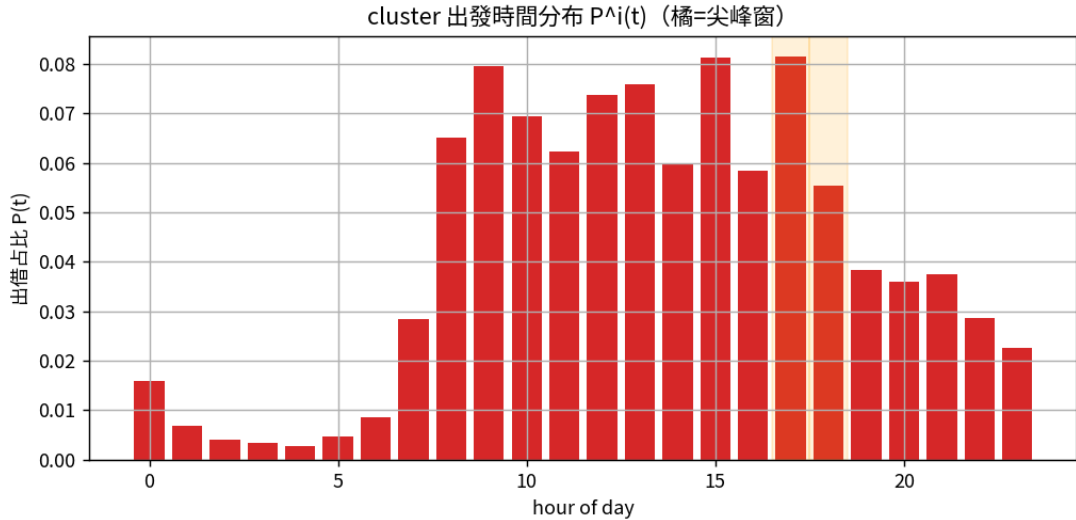


Figure 1: Cluster-aggregate departure distribution $P^i(t)$ (campus multi-peak; orange = peak window). The simulation uses each station’s own $P^i(t)$.

At the operating point ($\alpha \approx 0.48$, capacity = 100), the share of shortage users obtaining a bike rises from $\approx 6\%$ (without) to $\approx 15\%$ (with) information ($\approx 2.5\times$).

Mean cost without information is $\approx 10\text{--}11$ min, flat in α (random search is insensitive to shortage); with information it rises as the dispatch capacity saturates. The hypothesis holds: at 10–20% shortage information saves **2.4–3.2 min (22–31%)**; at the empirical $\approx 32\%$ it still saves ≈ 2.0 min (18%).

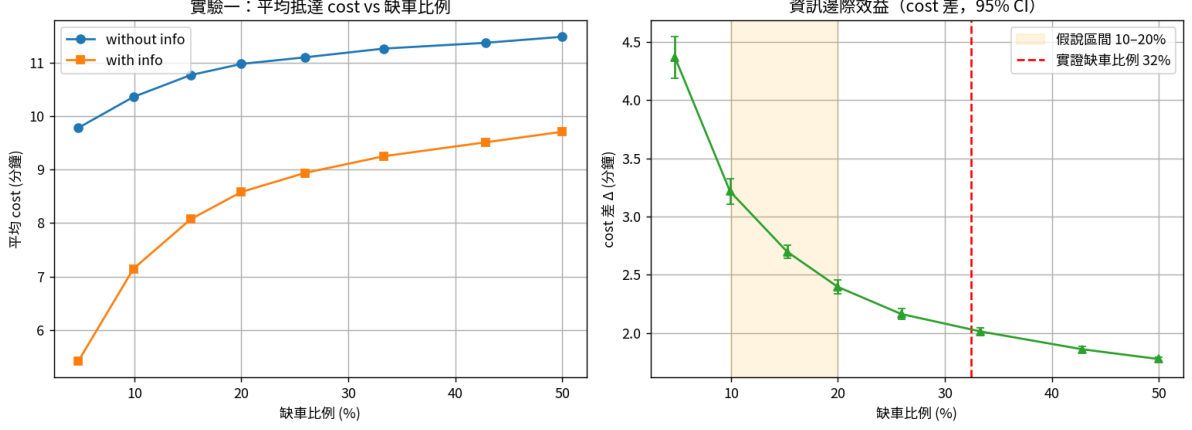


Figure 2: Exp. 1. Left: mean arrival cost per scenario. Right: cost difference Δ (95% CI); orange band = 10–20% hypothesis range, red dashed line = empirical $\approx 32\%$ shortage. Dispatch capacity = 100 bikes/window.

Table 2: Exp. 1 (capacity = 100 bikes/window).

Shortage	Without	With	Δ (saved)	Relative
4.8%	9.8 min	5.4 min	4.4 min	45%
9.9%	10.4	7.1	3.2	31%
15.3%	10.8	8.1	2.7	25%
20.0%	11.0	8.6	2.4	22%
25.9%	11.1	8.9	2.2	19%
33.3%	11.3	9.3	2.0	18%
50.0%	11.5	9.7	1.8	15%

Information is most valuable when shortage is low and dispatch ample (upper-left). Within the plausible band at the empirical shortage, the improvement is **15–22%** ($\Delta \approx 1.8\text{--}2.4$ min); raising dispatch from ≈ 50 to ≈ 400 bikes lifts it from $\approx 15\%$ to $\approx 29\%$.

Across the full τ range the curves differ by less than 0.5 min with no consistent direction: the dense real footprint makes the tolerance rarely bind, so the conclusion does not depend on τ .

When concentrated at a single station, the best locations lie in the **eastern dormitory/department area** (around Men’s Dormitory 1 and Mingda Hall), near the most shortage-prone interior stations (second EE building 62%, Forestry building 57%). The current main hub, MRT Gongguan, sits at the south-west edge and is sub-optimal for the campus shortage population. An agent-movement animation ([results/anim.html](#)) shows the two scenarios side by side on a shared clock.

8 Discussion

Why information helps: the dominant cost is walking far after failing to find a bike; information turns “gamble at the nearest station” into “head to a dispatch station that certainly has a bike within an acceptable detour,” replacing walking with riding. **Complementarity:** information needs sufficient dispatch to point users toward; under extreme shortage the capacity saturates and the marginal benefit falls. **Relation to prior work:** whereas others optimize the policy—limited in practice by manpower and vehicles [5]—we show that under the *same* constrained supply, disclosing existing locations already yields a sizeable, near-zero-cost saving (an app layer). **Policy:** for further gains, placing dispatch resources in the eastern campus area is more efficient than reinforcing the peripheral Gongguan hub.

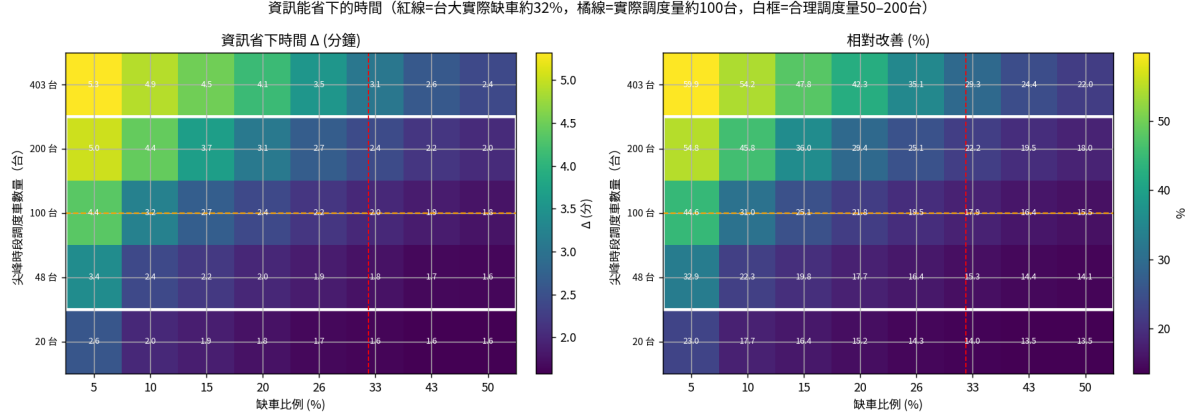


Figure 3: Δ (min) and relative improvement (%) over shortage $\times$ dispatch quantity. Red line = empirical shortage, orange line = mid-estimate 100 bikes, white box = plausible band 50–200.

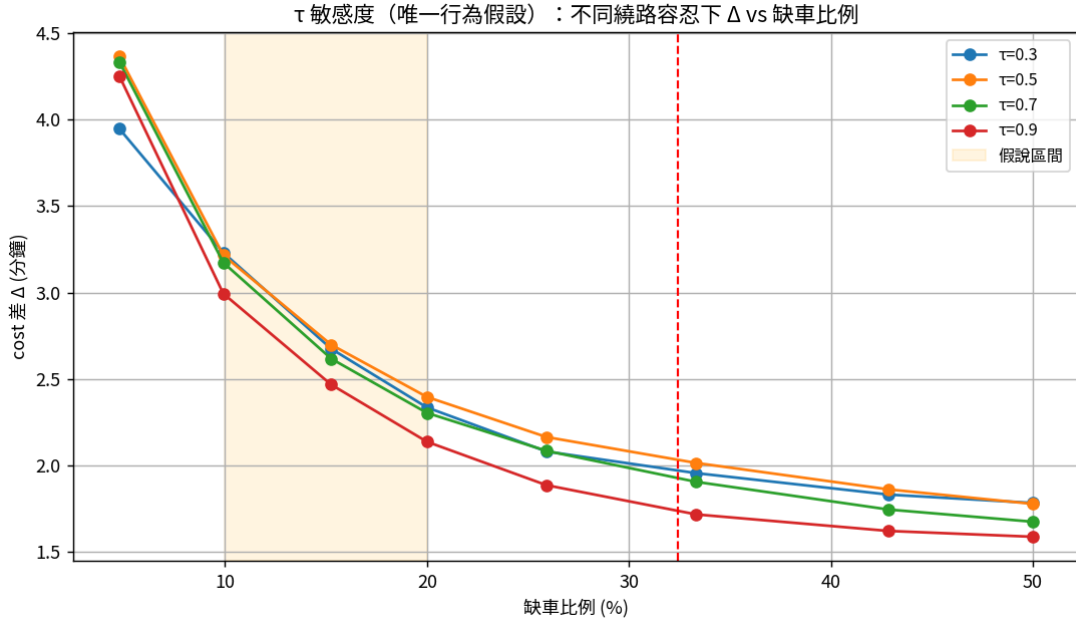


Figure 4: Δ vs shortage for $\tau \in \{0.3, 0.5, 0.7, 0.9\}$.

9 Limitations

Dispatch quantity is the largest uncertainty (proxy detection, no official records, possible return-burst inflation; hence anchored at a physical estimate and swept 50–200). Temporal mismatch: dynamics from Sept. 2025 vs. OD from Dec. 2025. Without information, users try only the single nearest station. Shortage is approximated by the `sbi` = 0 time fraction. Return cost is an expected-value approximation. Waiting, cars, and ride-sharing are not modeled.

10 Conclusion

Disclosing the locations of the existing dispatch bikes measurably shortens peak-hour arrival time on the NTU campus: at the empirical $\approx 32\%$ shortage across a plausible dispatch capacity (50–200 bikes/window) it saves ≈ 1.8 – 2.4 min (15–22%), and 2.4–3.2 min (22–31%) in the 10–20% range. The result is robust to the only behavioral assumption τ . Information and supply are complementary; given that real dispatch is resource-constrained, information disclosure is a low-cost lever that complements rather than replaces dispatch optimization. The study is

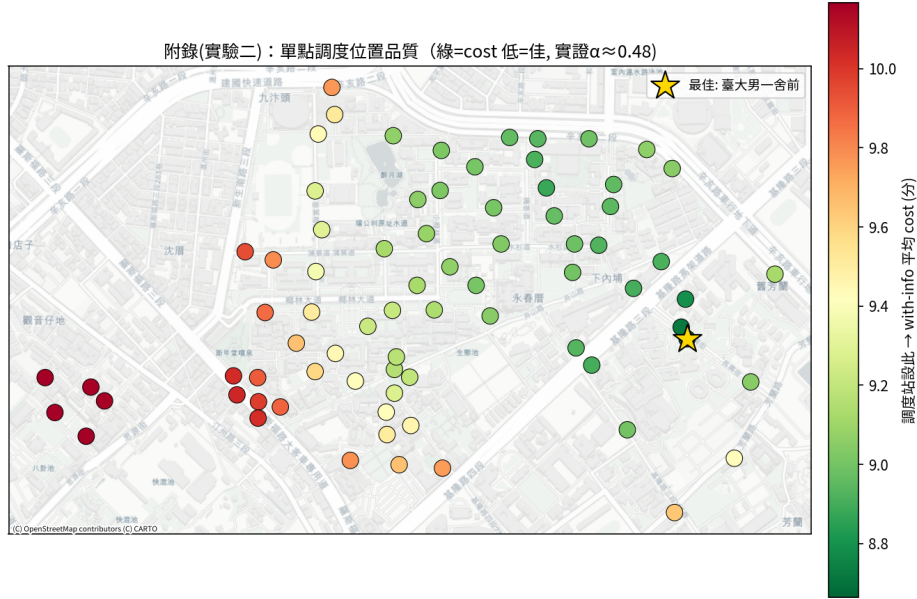


Figure 5: With-info mean cost when the total capacity is concentrated at each candidate station (green = better).

data-driven and openly reproducible.

References

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A Reproducibility

`pip install -r requirements.txt; bash fetch_data.sh; cd src && bash run_pipeline.sh.`
 Processed parquet files are committed, so the simulation and figures reproduce *without* the 44 GB raw data (`cd src && python experiments.py && analyze.py && viz_map.py && export_anim.py`).
 A fresh clone reproduces the reported numbers exactly (e.g., at 4.8% shortage $\Delta = 262\text{ s}/45\%$; best single location = Men’s Dormitory 1, 8.66 min).